2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT) | 979-8-3503-7024-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICCCNT61001.2024.10725616

Enhancing Diagnosis of Infectious Skin Disease: A Professional Approach using Deep Learning for Leprosy Detection

1st Chetan Jitendra Gawali
Department of Computer Science and Applications
Amrita School of Computing,
Amrita Vishwa Vidyapeetham, Amritapuri, India

2nd S. Subbulakshmi
Department of Computer Science and Applications,
Amrita School of Computing,
Amrita Vishwa Vidyapeetham, Amritapuri, India

***Abstract*—Leprosy remains a significant public health challenge, particularly in India. Early and accurate diagnosis is crucial for effective treatment and preventing nerve damage. This study explores the potential of deep learning techniques to improve leprosy diagnosis, specifically focusing on tuberculoid leprosy. Convolutional Neural Networks (CNNs) were investigated for leprosy detection using skin lesion images. A comparative analysis was conducted using VGG16, MobileNet v1, Xception v1 and EfficientNet B0 models.With a recall of 0.9298, an F1 score of 0.9217, and an accuracy of 0.9138, EfficientNetB0 performed the best. This indicates that EfficientNetB0 is suitable for leprosy diagnosis due to its ability to balance precision and recall. The study highlights the potential of deep learning for improving leprosy diagnosis and paves the way for more accurate, reliable, and accessible diagnostic tools.**

***Index Terms*—leprosy, EfficientNet, Xception, Convolutional Neural Networks, MobileNet**

I. Introduction

Leprosy, a chronic infectious disease caused by Mycobacterium leprae, remains a significant public health challenge, particularly in India. Despite advancements in national control programs, accurate and early diagnosis of leprosy continues to be hindered by several factors. The variable clinical presentation of leprosy, ranging from localized lesions in tuberculoid leprosy to widespread involvement in multibacillary leprosy, poses a challenge for healthcare providers, especially during the early stages of the disease. Traditional diagnostic methods, such as clinical examination and slit-skin smear microscopy, are subjective and have limitations in sensitivity. Slit-skin smear microscopy, while a mainstay for diagnosing tuberculoid leprosy, can miss paucibacillary cases with few lesions. Additionally, these methods rely heavily on the expertise of healthcare professionals, leading to potential inconsistencies in diagnosis, particularly in resource-limited settings with limited access to specialists. Delays in accurate diagnosis can have devastating consequences, including irreversible nerve damage and disability. Early and accurate identification of leprosy is crucial for effective treatment and preventing these complications [1].

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This study explores the potential of deep learning approaches to overcome the boundaries of existing diagnostic methods and improve their precision of leprosy testing, with a special focus on tuberculoid leprosy. Deep learning, an area of , has achieved exceptional success in a variety of image detection functions. Convolutional Neural Networks (CNNs), a special sort of deep learning architecture, have attained top-notch performance in image classification task. [2].

In this work, we propose a novel deep learning approach for diagnosing Tuberculoid leprosy, a challenging variant with localized lesions. We evaluate four pre-trained CNN models (VGG16, MobileNet, Xception, EfficientNet) and compare them to traditional classifiers. This comprehensive analysis highlights the advantages of deep learning, particularly EfficientNet B0's suitability for resource-constrained settings due to its efficiency. This paves the way for mobile-based leprosy diagnosis, potentially improving accessibility in remote areas.

The core objective of this research is to develop a robust and efficient system for leprosy diagnosis using CNNs. Utilizing transfer learning methods and pre-trained models, our goal is to diagnose leprosy with high accuracy and minimal computing complexity [3]. This is particularly important for potential real-world deployments in resource-constrained settings.

II. Related Work

In our study, we referred to the first scientific attempt at leprosy utilizing leprosy, which was done in 2016. Baweja and Parhar presented a convolutional neural network (CNN) architecture for leprosy lesion recognition [4]. Their work highlighted the significance of early diagnosis for preventing nerve damage and disabilities. They trained their CNN model on a combination of DermNetNZ datasets and web-scraped images, achieving an accuracy of 91.6%. However, the study focused solely on CNN-based lesion recognition without exploring other aspects of leprosy diagnosis or treatment. Additionally, the model's generalizability remains unclear as its performance was not evaluated on independent datasets or compared against existing methods.

Where in 2017, Kumar et al. proposed a CNN-based system for leprosy lesion detection utilizing the DermNetNZ dataset

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and web-scraped images [5]. Their model achieved an accuracy of 91.6%, highlighting the potential of deep learning for leprosy diagnosis. This study builds upon this work by focusing specifically on Tuberculoid leprosy, employing a pre-trained EfficientNet model for feature extraction, and investigating the use of SVMs for final classification [6]. In 2019 Velasco et al. developed a smartphone-based skin disease classification system utilizing the MobileNet convolutional neural network (CNN) model [7]. Focusing on Android application deployment, the researchers employed transfer learning techniques to train the model on a dataset encompassing seven distinct skin diseases: acne, eczema, pityriasis rosea, psoriasis, tinea corporis, varicella, and vitiligo. Throughout the investigation, the team explored various methodologies to optimize the accuracy of the MobileNet model. Initially, employing undersampling methods alongside default preprocessing techniques yielded a modest accuracy of 84.28%. Subsequently, utilizing an imbalanced dataset with default preprocessing significantly improved accuracy to 93.6%. Further enhancements were achieved through oversampling techniques, resulting in an accuracy of 91.8%. Ultimately, the combination of oversampling and data augmentation strategies yielded the highest accuracy of 94.4%. This research underscores the potential of deep learning and CNNs in developing robust classification systems for dermatological applications, offering valuable insights into methodologies for improving model accuracy in skin disease diagnosis on mobile platforms.

In a relevant study by De Souza et al. (2021) [8], a cross-platform mobile application for leprosy screening was developed utilizing machine learning. Their work employed a supervised learning approach with a Random Forest algorithm trained on a dataset of leprosy cases extracted from a Brazilian national database. The resulting model achieved satisfactory accuracy in classifying leprosy across different regions, demonstrating the potential of AI for leprosy diagnosis, particularly in remote areas. This work highlights the successful application of machine learning for pattern recognition in leprosy classification, paving the way for further exploration of deep learning techniques in this domain. A recent study by Baweja et al. (2023) [9] explored the potential of explainable AI for leprosy diagnosis. Their work introduced LeprosyNet, an AXI-CNN architecture, achieving high accuracy (98%) in leprosy prediction. The authors leveraged various explainable AI techniques, including Activation Layer Visualization, Occlusion Sensitivity, and Grad-CAM, to enhance model interpretability. Additionally, they compared LeprosyNet's performance to established models like AlexNet and ResNet, demonstrating its potential for accurate leprosy classification.

Athina et al. (2022) conducted a study comparing four deep learning models for detecting skin diseases, including ResNet50V2, VGG16, InceptionV3, and InceptionResNetV2 [10]. They utilized Explainable Artificial Intelligence (XAI) through LIME to enhance transparency in model predictions. A hybrid dataset containing diverse skin disease images was employed for training, ensuring robustness. Traditional classifiers were also compared, highlighting the superiority of deep learning. The InceptionResNetV2 model achieved a remarkable accuracy of 99.06%, showcasing its effectiveness in accurate diagnosis. Overall, the study underscores the potential of deep learning coupled with XAI for precise skin disease detection.

In 2023, the study done by Fernandes and his team [11] provides a valuable starting point by conducting a systematic literature review on the use of artificial intelligence (AI) in leprosy diagnosis. Their work highlights the potential of AI for improving diagnostic accuracy in dermatology, with a reported average accuracy of 89.97% across various skin diseases including leprosy. Notably, the review identifies limitations such as the restricted search scope and a scarcity of research specifically focused on leprosy diagnosis using deep learning techniques. This gap in knowledge underscores the need for further investigation into deep learning applications for leprosy detection, particularly for Tuberculoid leprosy, which is the focus of our current study.

Prior research explored deep learning for general leprosy detection. Our work tackles Tuberculoid leprosy, a more challenging variant. We compare deep learning models (including EfficientNet B0) to traditional methods, highlighting deep learning's advantages. EfficientNet B0's efficiency paves the way for mobile-based diagnosis in resource-limited settings. This focus on Tuberculoid leprosy, mobile deployment, and comprehensive analysis differentiates our research.

III. Solution Approach

The objective of this research is to design a reliable and effective leprosy diagnosis system by utilizing the pre-trained Convolutional Neural Network (CNN) models. The chosen models—VGG16, MobileNet v1, and Xception v1—are known for their high performance in image classification tasks. Additionally, EfficientNet B0 was selected for its optimized balance of accuracy and computational efficiency [12]. Utilizing pre-trained weights from the ImageNet dataset, these models make use of transfer learning to enhance feature extraction and ultimate performance of the model. Below, we outline the detailed solution approach, including the rationale for model selection, transfer learning techniques [13], and specific configurations used in this research.

1) Model Selection and Rationale: CNN models for this study have been selected depending on their architectural characteristics and proved competence when handling image classification tasks:

1) VGG16: Known for its simplicity and depth, the VGG16 model uses a straightforward approach with deep stacks of 3x3 convolutions, followed by fully connected layers. Despite its high parameter count, VGG16 provides a strong baseline for image classification.
2) MobileNet v1: Designed for efficiency, MobileNet v1 employs depthwise separable convolutions to reduce the number of parameters and computational load. This makes it suitable for applications requiring real-time performance on resource-constrained devices [14].

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3) Xception v1: This model extends the Inception architecture by replacing the standard modules with depthwise separable convolutions, enhancing efficiency and performance. Xception v1 is particularly powerful for capturing rich feature representations with fewer parameters.
4) EfficientNet B0: Leveraging a novel compound scaling method, EfficientNet B0 balances network depth, width, and resolution. This approach achieves high accuracy with fewer parameters and lower computational costs compared to traditional models, making it ideal for our task.

2) Transfer Learning and Model Adaptation: To optimize the performance of these pre-trained models for leprosy detection, we employed transfer learning techniques. Transfer learning allows us to utilize the knowledge gained from large datasets (ImageNet) to improve performance on our specific dataset with limited data.

- Pre-trained Weights: The convolutional bases of VGG16, MobileNet v1, Xception v1, and EfficientNet B0 were initialized with weights pre-trained on the ImageNet dataset. This step enables the models to leverage the rich feature representations learned from a diverse set of images.
- Input Shape: All models were adapted to accept input images resized to 128x128 pixels with three color channels (RGB), a standard format for image classification tasks.
- Excluding Top Layers: For each model, the fully connected (top) layers were excluded to allow for customization. This step involves adding task-specific layers that are tailored to the leprosy detection task.

IV. Architecture Diagram

As shown in Figure 1, this convolutional neural network model is design to diagnose leprosy by input images, using binary classification and hybrid model. It builds on pre-trained deep learning models i.e. VGG16, MobileNet v1, Xception and EfficientNet model from TensorFlow Hub, which serves as a powerful feature extractor. This means it takes advantage of knowledge gained from a vast dataset during pre-training. [15] After extracting features, the models processes them through additional dense layers to learn representations specific to the task at hand. To make the model more robust and improve its ability to generalise, various data augmentation techniques are applied to the training data. These include rescaling, zooming, rotation, and adjusting brightness [16]. The model is trained using the hinge loss function, which works effectively for problems related to binary classification, along with the Adam optimizer [17]. To prevent overfitting and ensure the best performance on validation data, regularisation techniques such as L1L2 regularisation and early stopping are used. Ultimately, the model outputs a single value indicating the likelihood of leprosy in the input image [18].

V. Methodology

This study aimed to diagnose leprosy using state-of-the-art Convolutional Neural Network (CNN) models, specifically

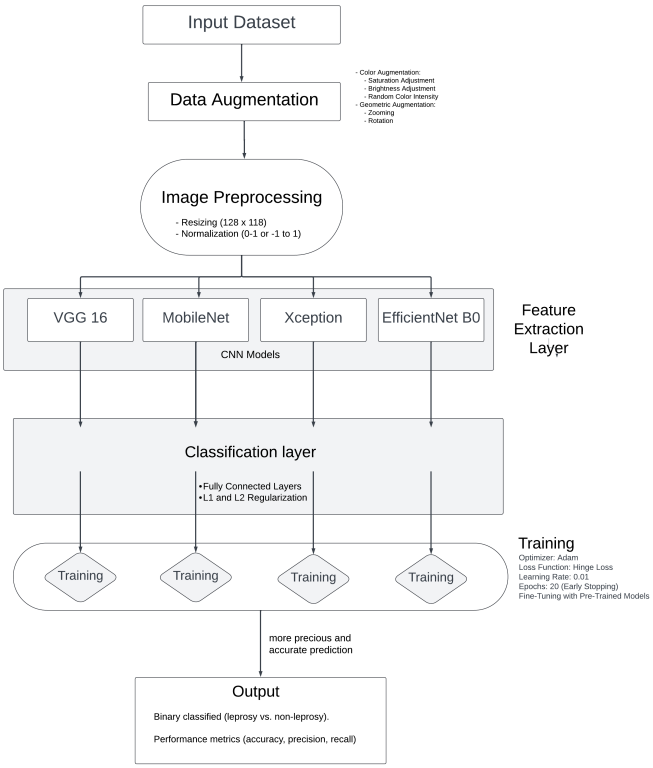


Fig. 1. Schematic architectural diagram of Proposed Methodology

VGG16, MobileNet v1, and Xception v1. These models were selected as to their demonstrated efficacy in image classification problems and their availability of pre-trained weights on the ImageNet dataset, which helps in transfer learning to leverage the learned features for our specific task.

A. Dataset

The dataset is publicly available on the open-source Kaggle platform, comprised a total of 1476 images. Figure 1 presents 15 example images from two classes of the utilized dataset. These images were specifically classified as tuberculoid leprosy, a subtype of the disease. To ensure robust model performance and prevent overfitting, the dataset is divided into three distinct parts: train (1476 images), valid (62 images), and test (62 images). The splitting technique enables the framework to learn from the training data, fine-tune hyperparameters on the validation set, and ultimately assess its generalizability on the unseen testing data [19]. (See sample of images from dataset in fig.2)

B. Data Augmentation

To increase dataset diversity and enhance the model's generalization capabilities, various data augmentation techniques were applied:

1) Color Augmentation::

- Saturation adjustment: Modifying the intensity of colors to simulate images taken under different lighting conditions.

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Fig. 2. Samples of Images with two classes

- Brightness adjustment: Changing image brightness to help the model recognize objects in diverse lighting scenarios.
- Random Color Intensity: Introducing random variations in color intensity to prevent the model from overfitting to specific color patterns.

2) Geometric Augmentation::

- Zooming: Applying zoom factors to alter the size of objects in the images, allowing the model to recognize objects at varying distances.
- Rotation: Rotating images by a certain degree range to help the model recognize objects from different angles.

C. Experimental Setup

This section details the experimentation conducted to evaluate different models for leprosy detection using image classification. The primary focus is on comparing feature extraction capabilities and classification performance of EfficientNet B0 against VGG16, MobileNet v1, and Xception v1. before feature extraction we will go with Pre-processing.

D. Data Preprocessing :

The raw leprosy images underwent various preprocessing steps to prepare them for model training:

Resizing: Images were resized to a standard dimension (e.g., 224x224 pixels) to ensure compatibility with the EfficientNet B0 model and efficient processing.

Normalization: Pixel values were normalized to a specific range (e.g., 0-1 or -1 to 1) for improved model convergence and training stability.

The core of the model is a Pre-trained EfficientNet B0 architecture obtained from TensorFlow Hub.

- Transfer Learning: EfficientNet B0 leverages knowledge learned from a vast image dataset (ImageNet) during pre-training. This pre-trained feature extractor helps the model learn robust features for leprosy classification, especially with a potentially limited leprosy image dataset. [20]
- Efficiency: EfficientNet B0 offers a good balance between accuracy and computational efficiency. This is crucial for real-world deployment scenarios where computational resources might be limited. [21]

E. Feature Extraction

All models leverage transfer learning, utilizing pre-trained weights from large datasets for feature extraction. Here's a breakdown of the pre-trained models and their feature extraction approaches:

- **VGG16:** This model uses the pre-trained VGG16 architecture with deep stacks of 3x3 convolutional layers. Despite its effectiveness, VGG16 has a high parameter count, potentially impacting efficiency during feature extraction.
- **MobileNet:** This model utilizes a pre-trained MobileNet v1 architecture, known for its efficiency in resource-constrained environments. MobileNet v1 employs depthwise separable convolutions for efficient feature extraction.
- **Xception v1:** This model leverages a pre-trained Xception v1 architecture, inspired by Inception, which utilizes depthwise separable convolutions for feature extraction. While offering good performance with large datasets, Xception can be computationally intensive.
- **EfficientNet B0:** This model utilizes a pre-trained EfficientNet B0 model from TensorFlow Hub as the feature extractor. EfficientNet employs compound scaling, which uniformly scales depth, width, and resolution to achieve optimal feature extraction capabilities.
  - First Dense Layer: This layer uses ReLU activation and contains 128 units. ReLU adds non-linearity to the extracted features, allowing the model to figure out complicated patterns [22]. The purpose of this layer is to learn task-specific, higher-level features relevant to leprosy detection [23].
  - Final Output Layer: This layer works well for binary classification (leprosy vs. non-leprosy) as it contains a single unit with a linear activation function [24]. L1 and L2 regularization are applied to this layer to prevent overfitting by penalizing large weights in the model.

    Here, L1 and L2 regularization are employed to prevent overfitting during model training. L1 regularization (Lasso) encourages sparsity by penalizing the absolute weights, promoting simpler models. L2 regularization (Ridge) penalizes large weights to prevent model complexity, enhancing generalization on unseen data. Together, they balance model complexity and performance, crucial for robust leprosy detection across diverse datasets.

extracted features from the models are represented as vectors and passed to the classification layer. Feature extraction

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occurs at multiple levels: initial levels capture basic features like edges and corners, intermediate levels focus on textures and shapes, and advanced levels extract high-dimensional, disease-specific features crucial for leprosy detection using classifiers.

F. Classification

- **VGG16:** Similar to EfficientNet B0, VGG16 likely employs additional fully-connected layers for classification. However, due to its deep architecture, VGG16 might require careful hyperparameter tuning to prevent overfitting during the classification stage.
- **MobileNet:** MobileNet likely uses additional fully-connected layers for classification. Due to its focus on efficiency, the number of layers may be limited compared to other models.
- **Xception:** Xception might employ additional fully-connected layers for classification. Similar to VGG16, its complex architecture could necessitate hyperparameter optimization to avoid overfitting during classification.
- **EfficientNet B0:** This model adds two fully-connected (dense) layers after the pre-trained EfficientNet B0 for classification. The first layer learns higher-level features specific to leprosy detection, while the final layer with a single unit and linear activation performs binary classification (leprosy vs. non-leprosy). L1 and L2 regularization are applied to prevent verfitting.
  Following the pre-trained EfficientNet B0 layers, two fully-connected (dense) layers are employed for classification:
  - First Dense Layer: This layer has 128 units and utilizes ReLU activation. ReLU introduces non-linearity, enabling the model to capture complex patterns within the extracted features [22]. The purpose of this layer is to learn task-specific, higher-level features relevant to leprosy detection [23].
  - Final Output Layer: This layer has a single unit with a linear activation function, suitable for binary classification (leprosy vs. non-leprosy) [24]. L1 and L2 regularization are applied to this layer to prevent overfitting by penalizing large weights in the model.

G. Training

The training process involved optimizing the model's parameters to achieve the best possible performance on the leprosy classification task. Here's a breakdown of the training details:

- Optimizer: The Adam optimizer is used for its fast convergence compared to traditional gradient descent methods.
- Loss Function: The hinge loss function, suitable for binary classification and margin maximization, is employed.
- Learning Rate: A learning rate of 0.01 is chosen to strike a balance between convergence speed and training stability.
- Epochs: The each model is trained for a 20 of epochs, achieving a certain validation accuracy or some utilizing early stopping. Due to limited computational resources, 20 epochs were chosen as the maximum training duration. While less than 20 epochs could lead to underfitting, also all models showed signs of overfitting after 20 epochs. So, In this experiment 20 epochs offered a balance between sufficient learning and preventing overfitting.

Here, we have implemented early stopping to prevent overfitting. During training, if the validation accuracy doesn't improve for a certain number of epochs (20 in this case), the training is stopped early. The model with the best validation accuracy is restored, ensuring that the model doesn't continue training on data that may lead to overfitting.

VI. Results

The experimentation involved training and evaluating four different convolutional neural network (CNN) models for the binary classification task of leprosy detection. The models tested were EfficientNetB0, VGG16, MobileNet v1, and Xception v1. Each model was fine-tuned and evaluated using a consistent dataset, and their performance was assessed based on precision, recall, and F1 score. This section details the results of the experimentation and provides a comparative analysis of the models' performance.

Model	Precision	Recall	F1 Score
VGG16	0.7714	0.9474	0.8504
MobileNet V1	0.9592	0.8246	0.8868
Xception	0.8947	0.8947	0.8947
EfficientNet B0	0.9138	0.9298	0.9217

TABLE I
PERFORMANCE MATRIX OF EACH MODEL

A. Comparative Analysis

In this Section, We are doing comparative analysis of all selected CNN models, i.e. VGG16, MobileNet v1, Xception v1 and EfficientNet B0.(See Table no.II)

1) VGG16: VGG16 exhibited a high recall (0.9474), which means it identified almost all positive cases of leprosy. However, it had the lowest precision (0.7714) among the models, indicating a higher rate of false positives. Consequently, its F1 score was lower (0.8504) than EfficientNetB0 and Xception. The VGG model achieved a training accuracy of 0.98 and a validation accuracy of 0.95, indicating good performance but also potential signs of overfitting, as indicated by the lower precision.

2) MobileNet v1: MobileNet v1 showed the highest precision (0.9592), indicating it had the fewest false positive predictions. However, its recall (0.8246) was the lowest among the models, meaning it missed more positive cases of leprosy. The F1 score (0.8868) reflects this trade-off. MobileNet v1's design, focused on efficiency using depthwise separable convolutions, made it effective in minimizing false positives but less effective in identifying all positive cases. In result, the model achieved a training accuracy of 0.95 and a validation accuracy

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of 0.85, indicating good performance but with potential signs of overfitting.

3) Xception v1: Xception v1 achieved a balanced performance with both precision and recall at 0.8947, resulting in an F1 score of 0.8947. This model's use of depthwise separable convolutions, inspired by the Inception architecture, allowed it to perform well across both metrics, though not as outstanding as EfficientNetB0. The Xception model achieved a high training accuracy of around 0.90 but a lower validation accuracy of 0.70, indicating potential overfitting.

4) EfficientNetB0: EfficientNetB0 achieved the highest F1 score (0.9217), indicating a well-balanced performance between precision and recall. This model effectively captured positive leprosy cases (high recall of 0.9298) while maintaining a relatively low rate of false positives (high precision of 0.9138). The compound scaling approach used in EfficientNetB0, which balances depth, width, and resolution, likely contributed to its superior performance. While training, The model achieved a training accuracy of 0.95 and a validation accuracy of 0.90 after 15 epochs, indicating good performance with signs of potential overfitting. EfficientNetB0's use of MBConv blocks and the Swish activation function provided a robust feature extraction capability, making it highly suitable for the leprosy detection task.

Model	Train Accuracy	Val Accuracy	Trainable Params
VGG 16	0.98	0.95	50343937
MobileNet	0.95	0.85	1050625
Xception	0.9	0.7	2099201
EfficientNet B0	0.95	0.9	164097

TABLE II
TRAINING ACCURACY AND VALIDATION ACCURACY

Overall, after performing comparative analysis and evaluating four deep learning models for leprosy detection, EfficientNetB0 achieved the highest F1-score (0.9217), indicating a superior balance between precision (0.9138) and recall (0.9298) compared to VGG16 (high recall but low precision), MobileNet v1 (high precision but low recall), and Xception v1 (moderate precision and recall). This suggests EfficientNetB0's compound scaling approach, MBConv blocks, and Swish activation function effectively extract features for accurate leprosy detection with minimal false positives, despite all models exhibiting signs of overfitting.

VII. CONCLUSION

Leprosy remains a significant public health challenge, particularly in India. This research explored the potential of deep learning for improving leprosy diagnosis, specifically focusing on Tuberculoid leprosy. Traditional diagnostic methods for leprosy, such as clinical examination and slit-skin smear microscopy, face numerous limitations including variability in clinical presentation, reliance on expert interpretation, and low sensitivity in early-stage detection. These challenges underscore the urgent need for innovative diagnostic approaches that can offer greater accuracy and consistency.

Our research explores the use of deep learning models, specifically leveraging the EfficientNetB0 architecture, for the diagnosis of tuberculoid leprosy. By utilizing a pre-trained EfficientNetB0 model for feature extraction and employing support vector machines (SVMs) for final classification, we aim to enhance the accuracy and robustness of leprosy detection. This approach is supplemented with various data augmentation techniques to improve the model's generalizability and performance.

In our comparative analysis, the EfficientNetB0 model demonstrated superior performance with a precision of 0.9138, recall of 0.9298, and an F1 score of 0.9217, outperforming other models like VGG16, MobileNet v1, and Xception v1. This indicates that EfficientNetB0 strikes an optimal balance between precision and recall, making it highly suitable for leprosy diagnosis. The model's high precision and recall suggest its potential in effectively identifying positive leprosy cases while minimizing false positives, which is crucial for timely and accurate treatment.

The study contributes to the existing body of work by addressing the specific challenges of diagnosing tuberculoid leprosy and proposing a deep learning-based solution that can be deployed in resource-constrained settings. The use of EfficientNetB0, along with SVMs for classification, represents a significant advancement in the field of leprosy diagnosis, offering a promising tool for healthcare providers.

However, further research is needed to address limitations. Data imbalance, where leprosy images might be fewer, needs to be tackled with techniques like oversampling or undersampling. External validation on an independent dataset is crucial to ensure the model's generalizability in real-world scenarios. Additionally, while the model performs well, understanding its decision-making process for improved interpretability would be valuable.

Future work will focus on further validating this model on larger and more diverse datasets, exploring the integration of explainable AI techniques to enhance interpretability, and investigating the potential for real-world deployment through mobile health applications. This research paves the way for more accurate, reliable, and accessible diagnostic tools for leprosy, ultimately contributing to better management and eradication efforts for this enduring disease.

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